

AYMANE AIT DADS

Applied Machine Learning Intern | Audio & Time-Series Modeling · Industrial AI · Data Quality & Evaluation

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by designing and owning an end-to-end LoRA SFT pipeline on a 30B-parameter model - demonstrating the ability to develop, evaluate, and debug large-scale ML systems under real constraints. At Orange Maroc, processed and structured 100K+ fiber-optic network records, building automated feature engineering pipelines that reduced manual analysis time by 60% and delivered clustering outputs directly used by the network optimization team. Bringing hands-on experience in time-series and audio signal modeling (Mel spectrograms, SpecAugment, transformer autoencoders), data validation, ablation-driven evaluation, and end-to-end model ownership using PyTorch and Scikit-learn - exactly the applied skill set WaveWorks needs for live industrial deployment. Eager to apply rigorous data quality validation and modeling methodology to WaveWorks' real-world industrial audio environments and contribute directly to production-ready applied AI systems.

CORE COMPETENCIES

Large-Scale Dataset Processing | Data Quality Validation | Audio & Time-Series Modeling | Model Development & Evaluation | Anomaly Detection | Ground Truth Alignment | Ablation Studies | Industrial AI Deployment

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Processed, cleaned, and structured **100K+ fiber-optic network records** (upload speed, latency, jitter), performing ground truth alignment and data quality validation across thousands of network nodes.
- Developed a clustering pipeline using K-Means that mapped customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for maintenance decisions.
- Built an automated feature engineering and model evaluation workflow using Random Forest classifiers, reducing **manual analysis time by ~60%** and enabling repeatable, scalable network quality classification.
- Delivered stakeholder presentations translating model outputs into actionable maintenance recommendations, bridging data science findings with **operational decision-making** for a non-technical audience.

PROJECTS

Anomalous Sound Detection - Industrial Equipment

EURECOM · 2025 · AUC 0.80 Predictive Maintenance

- Designed an unsupervised fault detection system for slide-rail industrial machinery using a transformer autoencoder on **Mel spectrograms with SpecAugment**, achieving AUC 0.80 with no labeled anomaly data available.
- Built the full audio data processing pipeline - framing, spectrogram extraction, augmentation, and reconstruction-error evaluation - directly mirroring the **large-scale time-series dataset structuring** and model evaluation workflow required for live industrial deployment.

PyTorch · Mel Spectrograms · SpecAugment · Transformer Autoencoder · Scikit-learn

NVIDIA Nemotron Model Reasoning Challenge

Kaggle · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context), owning the full pipeline from data preparation through evaluation.
- Ran **controlled ablation experiments** across learning rate schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT - and built custom cipher parsers and 8-bit operation solvers to produce verified chain-of-thought training data.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Statistics, Image Security, Cloud Computing, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & Engineering: PyTorch | Scikit-learn | TensorFlow | Hugging Face Transformers | Unsloth | TRL | PEFT | vLLM

Data Processing & Modeling: Pandas | NumPy | Feature Engineering | Time-Series Analysis | Mel Spectrograms | Clustering | Ensemble Methods | EDA

MLOps & Infrastructure: Python | SQL | Git | Linux | FP16/BF16 Mixed Precision | Ablation Studies | Power BI | Docker (basic)